REHEARSALS

AI-Powered Conversation Practice Tool

Project Abstract — June 2026

OVERVIEW

Overview

Rehearsals is a single-page web application that enables users to practice high-stakes personal conversations before they happen in real life. The core premise is simple: describe a person you need to talk to and what makes that conversation hard, and the AI will simulate that person realistically so you can rehearse your approach in a safe, private environment. After the rehearsal ends, the app delivers an AI-generated coaching debrief with structured feedback and an emotional heatmap of the conversation.

The application targets anyone facing a difficult interpersonal conversation — whether confronting a parent about a sensitive topic, navigating a performance review with a defensive manager, or setting boundaries with a roommate. By lowering the cost of practice, Rehearsals helps users enter real conversations with greater clarity, confidence, and emotional preparedness.

TECHNICAL ARCHITECTURE

Technical Architecture

Rehearsals is built as a zero-dependency, client-side HTML application. The entire product — UI, logic, and API integration — lives in a single index.html file, making it instantly deployable with no build step, no server, and no backend. The application communicates with the Groq API using the Llama 3.3 70B model, chosen for its strong instruction-following, naturalistic dialogue generation, and free-tier availability under Groq's inference platform.

The UI is structured as three sequential screens: a Setup screen where the user configures their persona, a Chat screen for the live rehearsal, and a Debrief screen displaying post-session coaching. State is managed in a plain JavaScript object. Voice input is handled via the Web Speech API (webkitSpeechRecognition), and text-to-speech output is delivered through the browser's SpeechSynthesis API, enabling fully hands-free conversation practice.

KEY FEATURES

Key Features

Persona Engine: Users describe who they need to talk to, what makes that person difficult, and what they want to accomplish. A structured system prompt instructs the LLM to embody this person realistically — not as a caricature, but with human nuance, defenses, and occasional openness.

Live Rehearsal Chat: A polished chat interface with typing indicators, auto-resizing input, and smooth message animations. The persona responds in short, naturalistic turns to mirror real conversation pacing.

Voice I/O: Users can speak their side of the conversation via microphone, and the persona responds audibly via text-to-speech, enabling immersive spoken practice without typing.

AI Coaching Debrief: After ending the rehearsal, a second LLM call analyzes the full transcript and returns structured JSON feedback: what the user did well, where they could hold their ground, and one specific alternative phrasing to try.

Emotional Heatmap: The debrief includes a segment-by-segment heatmap classifying each user turn as Comfortable, Hesitant, Defensive, or Shut Down, plus a one-sentence insight about the conversation's overall emotional arc.

DESIGN PHILOSOPHY

Design Philosophy

The visual design prioritizes focus and emotional calm: a dark-mode palette with deep neutrals and a single purple accent color minimizes distraction during what can be an emotionally charged activity. Typography is set in system-ui sans-serif at generous line-heights to reduce cognitive load. Every screen is intentionally narrow and vertically scrollable, ensuring the interface never feels overwhelming on either desktop or mobile.

Feedback in the Debrief screen is framed around three questions: what worked, where the user could have been stronger, and what to try differently. This structure is deliberately non-judgmental and action-oriented, drawing from principles of communication coaching rather than performance review. The heatmap provides an at-a-glance emotional narrative the user can internalize and act on.

LIMITATIONS & FUTURE WORK

Limitations & Future Work

The current implementation stores the API key in client-side JavaScript state, which is appropriate for a local or personal deployment but would require a backend proxy for any public multi-user release. Conversation history is held only in memory and is lost on page refresh; a future version could persist sessions to localStorage or a lightweight backend.

Planned enhancements include session history with replay, shareable debrief reports, persona presets for common archetypes (critical parent, dismissive manager, conflict-avoidant partner), and fine-tuned prompting for specific conversation types such as boundary-setting, salary negotiation, and conflict resolution. Streaming API integration would also improve perceived latency during persona responses.

CONCLUSION

Conclusion

Rehearsals demonstrates that a single HTML file, paired with a capable LLM backend, can deliver a meaningful, emotionally intelligent product. By making conversation practice accessible with no installation, no account, and no cost, it removes the barriers that prevent most people from preparing for the conversations that matter most. The combination of realistic persona simulation, multimodal voice interaction, and structured AI coaching creates a practice loop that is both practical and genuinely useful.

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